

Development of a Cylindrical Velostat-Driven Tactile Sensing System for Physical Interaction with a Social Robot

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Abstract— Social Assistive Robots (SARs) are increasingly explored as learning companions within primary school environments, offering new ways to engage pupils through embodied interaction, expressive behaviour, and natural communication. The ability to perceive physical contact and estimate interaction forces is essential to ensure safety and correct SARs' behaviours. This work presents the development and preliminary assessment of a cylindrical Velostat-driven sensor. The proposed sensing cylinder was designed to be easily integrated into the CASTOR robot's arm. The contact area and differentiation between contact forces were assessed with 6 types of pHRI. Five participants (38.40 ± 15.79 years old) tested the sensor integrated into the robot in 5 trials of randomized interactions. The outcomes demonstrate the feasibility of using a low-cost cylindrical tactile sensor to classify typically maintained physical Human-Robot Interactions (pHRIs) in a social robot with an accuracy of 84.67% across the classification types, showing confusion between pet and touch, and combining hand squeeze and handshaking, as these are similar interactions. Future work will implement the sensing capabilities of the robot's arm in activities with children, capturing data of real-world pHRIs.

I. INTRODUCTION

Socially assistive robots (SARs) have been increasingly studied as tools for primary education, offering new forms of engagement through physical presence, speech, gesture, and expressive behaviour [1]. Unlike digital media, SARs provide embodied interactions that can enhance motivation, sustain attention, and support participation across a variety of learning activities, including storytelling, literacy, and phonological awareness. Robots such as NAO [2], Pepper [3], and Cozmo [4] have been trialled in classrooms to model behaviours, demonstrate tasks, and act as peer-like learning companions.

Physical human-robot interaction (pHRI) is a defining feature of SARs operating in therapeutic, educational, and domestic environments [5]. In these contexts, the ability to perceive physical contact and estimate interaction forces and types is essential to ensure safety and correct robot behaviour. In addition, children often explore and communicate through gestures and contact [6]. Tactile sensors, soft surfaces, and huggable compo-

nents can enhance comfort and playful exploration.

Existing tactile sensing solutions often rely on different concepts in planar sensor arrays, custom taxel skins, or commercial force sensors [7]. Moreover, those sensors can be expensive, difficult to scale, or poorly suited to non-planar geometries [8]. Also, there are some issues in signal noise, baseline drift, and durability [9], which can limit the adoption of tactile systems in classroom-ready robots. In particular, due to their relevance for robotic and pHRI applications, cylindrical and non-planar shapes sensing configurations have been explored and validated in terms of force differentiation and classification [10], [11].

Materials such as Velostat have been used for distributed touch sensing, due to their flexibility, low cost, and ease of fabrication, enabling robots to detect interactions, objects, and specific patterns and respond appropriately [8]. Its pressure-dependent resistive properties allow the development of compliant and flexible sensing surfaces that can conform to complex geometries. Nevertheless, the integration of Velostat-based sensors into cylindrical structures suitable for robot arms requires careful design and validation to ensure reliable contact measurement across different pHRIs.

The application of tactile and pressure sensors for pHRI monitoring has been studied in a variety of scenarios. In prior work by Gaitán-Padilla *et al.*, a Velostat sensor enabled the CASTOR robot to generate more natural hugging interactions than approaches based only on predefined timing [12]. In parallel, sensor data have been used to train machine learning (ML) models that classify interaction states, including systems that recognize gestures occurring during hugs in autonomous huggable robots [13] and with small social robots to pet [14]. Such approaches enhance personalization and demonstrate improved social adaptability. However, achieving reliable estimation of more intricate pHRI, especially those involving detailed anatomical areas like the hand, still demands sensors with greater sensitivity and sensing areas [15], [16].

Regarding classification algorithms applied to Velostat-based sensors for pressure and force detection, several approaches have been reported in the literature. These include Convolutional Neural Networks (CNNs) [8], Random Forest

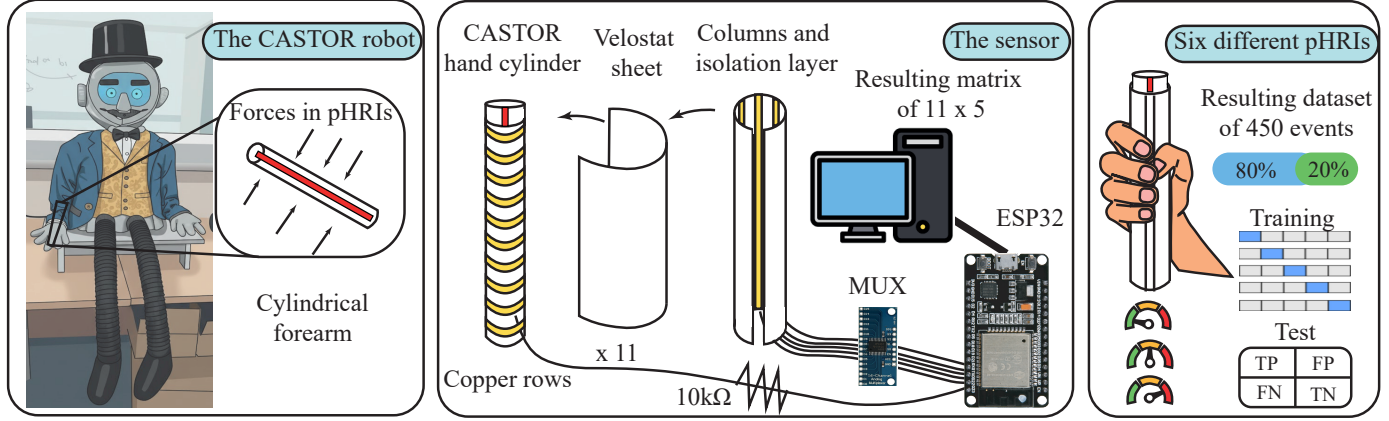


Fig. 1: Framework for the contribution stages of this work to classify pHRIs in the CASTOR robot, the developed sensor, and the protocol and ML classification process.

(RF) methods [17], and other Neural Network (NN) architectures [18]. Such algorithms have been used to identify pressure points, estimate force distributions, perform object detection over compliant surfaces, and monitor activities of daily living during therapy. While these methods demonstrate promising classification performance, they present different trade-offs in terms of accuracy, computational complexity, and suitability for real-time implementation on embedded microcontroller platforms, which has been evaluated and discussed [8].

The CASTOR Project is an initiative focused on the deployment of low-cost, modular social robots for therapeutic and educational applications involving children [19]. The project follows a participatory and therapist-informed design approach, aiming to balance accessibility, safety, and adaptability in real-world interaction scenarios. For this reason, the project seeks to continue deploying low-cost and co-design technology for its implementation at schools and therapy centres.

This work presents the development and experimental evaluation of a Velostat-based tactile sensing system integrated into the CASTOR social robot's hand structure. A controlled interaction protocol was designed in which participants performed repeated randomized trials of six types of pHRIs. The resulting dataset was used to train an RF classifier capable of distinguishing between interaction types such as petting, touch, and squeeze. The framework of the context, sensing architecture, and classification process is illustrated in Figure 1.

II. METHODOLOGY

This section describes the Velostat-based cylindrical sensing system, the procedure of the controlled interaction protocol, and the data management for the pHRI classification ML algorithm.

A. Cylindrical Sensor Development

The tactile sensing system was designed as a low-cost, compliant cylindrical module intended for direct integration onto the robot arm. The sensing concept is based on the pressure-dependent resistive properties of Velostat (3M, United States), which changes electrical resistance when subjected to mechanical deformation. This enables the estimation of contact intensity through variations in measured voltage through a voltage divider, making it suitable for force-aware pHRI.

Unlike traditional planar tactile arrays, the proposed sensor adopts a cylindrical geometry to better match the shape of the robot arm, where physical interaction commonly occurs [20]. A cylindrical configuration increases surface contact coverage, reduces dead sensing zones, and allows force detection regardless of direction of contact around the arm. This design also simplifies mechanical integration, as the sensor can be mounted as a sleeve-like structure without altering the robot's kinematics.

The sensor consists of a multilayer structure composed of conductive copper tape, a Velostat sensing layer, and an outer layer made of insulation tape. When external forces are applied to the surface, localized compression of the Velostat layer reduces resistance at the corresponding sensing location, enabling spatial and intensity-based contact detection. The resulting matrix was of 11 x 5 sensing points of 6.5 mm, providing 55 individual points distributed around the cylindrical surface.

B. Participants

Five adult volunteers (4 male, 1 female; mean age 38.40 ± 15.79 years) participated in the study. Participants were recruited from laboratory members at the University of the West of England (UWE) and took part on a voluntary basis. The inclusion criterion required participants to be adults with no

known motor impairments affecting hand function or grip ability. The experimental protocol was reviewed and approved by the institutional ethics committee.

C. Procedure

The researcher explained the experimental procedure, detailing the purpose of the interaction tasks, the types of arm interactions to be performed, the timing requirements, and the rest intervals between actions. Participants were allowed to ask clarification questions before the trials started. Each participant completed five trials of pHRI with the robot's arm integrated into the real position in the robot's arm, as shown in Figure 2. The six interaction types were hand touch, forearm touch, hand squeeze, forearm squeeze, handshake, and pet.



Fig. 2: Setup for the pHRI protocol showing the participant sitting in front of the CASTOR robot and the instrumented hand.

Each trial comprised 18 individual interactions, including three repetitions of each of the six defined types. The order of interactions within each trial was randomized to reduce ordering effects, and it was divided into five trials to reduce cognitive load, ensuring that all the trials had less than 2 minutes. For each interaction, participants performed the instructed contact action with the robot's arm following the researcher's verbal cue. A neutral position was needed between consecutive interactions to ensure signal separation and consistency.

C1 Data Acquisition

Data acquisition is handled by an ESP32 microcontroller (Espressif Systems, China). To reduce the number of required

pins in the final version (two instrumented arms with 5 columns each), the lines were connected to a multiplexer (Texas Instruments, United States), while rows were connected to 11 GPIO pins of the ESP32 with a 10 k Ω resistor for the voltage divider. The ESP32 scans the matrix by activating one row at a time and multiplexing the read column, reconstructing a full tactile frame at each acquisition cycle. The collected data points are transmitted via USB serial communication (at 25 Hz) to a host computer for visualization and analysis. On the computer, the incoming sensor values are mapped into a 11 x 5 grid visualized as a heatmap, where color represents normalized contact intensity.

In total, each participant performed 90 interactions. With five participants, the resulting dataset consisted of 450 recorded interactions, corresponding to 75 samples per interaction type. For the dataset, the saved data were divided and labeled by event. Initially, the 55-channel signal was segmented into individual interaction events using the Sum of Absolute Differences (SAD) and a global activity threshold. A second-order Butterworth low-pass filter with a 4 Hz cutoff was applied to each event. Heatmaps were generated to visualize the spatial force distribution in each interaction type.

Based on the heatmaps, touches and hand compression interactions were combined for a second classification algorithm, obtaining a four-option algorithm. The features for the training encompass statistical (mean, RMS, and peaks per channel), temporal (signal entropy and SAD), and spatial (Center of Pressure (CoP) and contact area) characteristics. An RF algorithm of a maximum of 100 trees was used; the model's performance was evaluated using 10-fold cross-validation, and feature importance analysis was conducted to identify the most significant tactile descriptors for distinguishing the interaction types.

III. RESULTS

This section presents the experimental results obtained of the development and assessment of the sensing cylinder in the pHRI classification. Figure 3 presents the heatmaps obtained from the Velostat-based sensor integrated into the robot's arm. The heatmaps represent the contact intensity along the sensor, where the beginning of the map corresponds to the initial region of the sensor positioned on the robot's arm. In the no-touch scenario, the heatmap displays values close to zero across the entire sensor matrix. This behavior indicates a low noise level and good signal stability, confirming that the system does not produce significant activations when no interaction occurs.

During tactile interactions, the sensor successfully distinguished between contact location and applied force. Hand-touch events produced low-to-moderate intensity activations localized at the proximal end of the heatmap, consistent with the

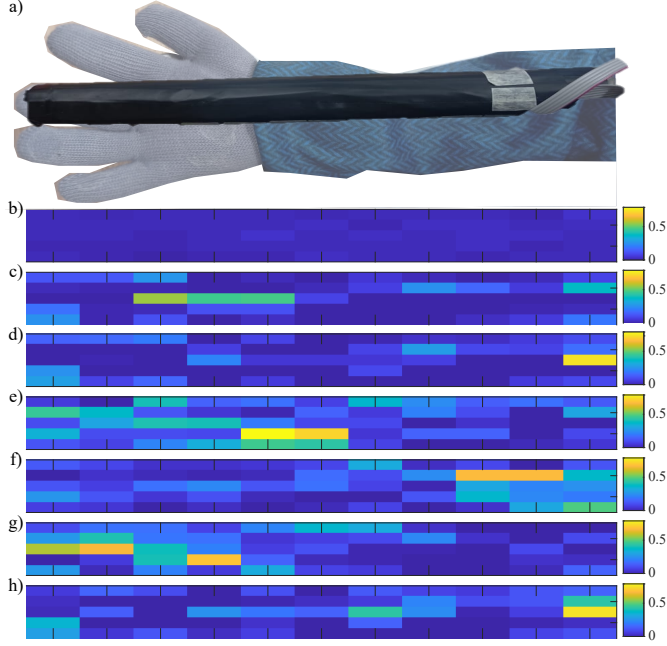


Fig. 3: a) The soft robot's hand with the sensor in the cylindrical shape. Example heatmaps of each interaction: b) no-touch, c) hand-touch, d) forearm-touch, e) hand-squeeze, f) forearm-squeeze, g) handshake movement, and h) pet, with the normalized intensity variation.

punctual application of low force. Conversely, forearm-touch interactions resulted in a distinct longitudinal shift in activation along the sensor array. While the sensor demonstrated high spatial sensitivity, significant intersubject variability was observed regarding the perceived anatomical boundary between the hand and forearm. To account for this subjective positioning and ensure classification robustness, these interactions were aggregated into a single "Touch" category for a second classification algorithm.

Compression interactions (squeezing) exhibited spatial patterns similar to touches but with significantly higher signal magnitudes. The hand-squeeze produced intense, localized activation, while the forearm-squeeze yielded both higher intensity values and a broader activation area, reflecting the increased contact surface of the forearm. Due to morphological similarities in the signal profiles of handshakes and hand-squeezes, these were also merged for the second classification task. Collectively, these data confirm the sensor's dual capacity for spatial localization and force-magnitude discrimination across diverse interaction types.

The initial RF model, trained on the original six interaction classes, yielded an overall accuracy of 72.23%. As illustrated in the confusion matrix (Figure 4), significant misclassifications occurred between the hand-squeeze and handshake categories (3 and 5), as well as between touches and pet interactions (1-2,

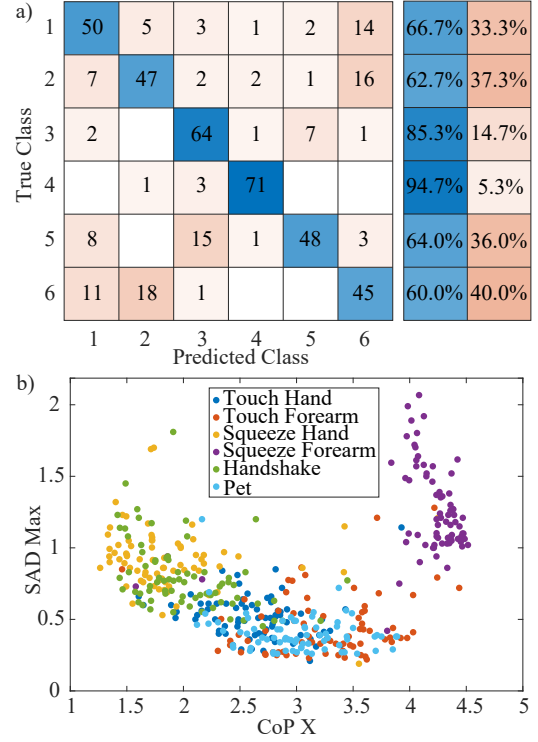


Fig. 4: Classification. a) Confusion matrix of the cross-validation with $k = 10$. b) Scatter of the two top features.

6), with the pets exhibiting the lowest individual class accuracy.

To improve classification performance, a second RF model was implemented by combining morphologically similar interactions, resulting in an increased overall accuracy of 84.67%. In this optimized model, the forearm-squeeze achieved the highest class-specific accuracy (94.7%), while pet interactions remained the primary source of error, frequently confounded with touches due to their overlapping low-intensity profiles (Figure 5a). Feature analysis (Figure 5b) underscores the critical role of the SAD and CoP characteristics. Because SAD-based features represent the temporal derivative of the sensor matrix, they capture the temporal dynamics necessary to distinguish the rhythmic motion of a pet from a static touch. Simultaneously, the CoP allows the model to determine the spatial centroid of the interaction, effectively localizing the contact point. Furthermore, the maximum values of the SAD features serve as primary discriminators for high-force, high-velocity interactions such as squeezes and handshakes.

IV. DISCUSSION

The heatmaps presented in Figure 3 illustrate the spatial distribution of sensor activations across the cylindrical forearm,

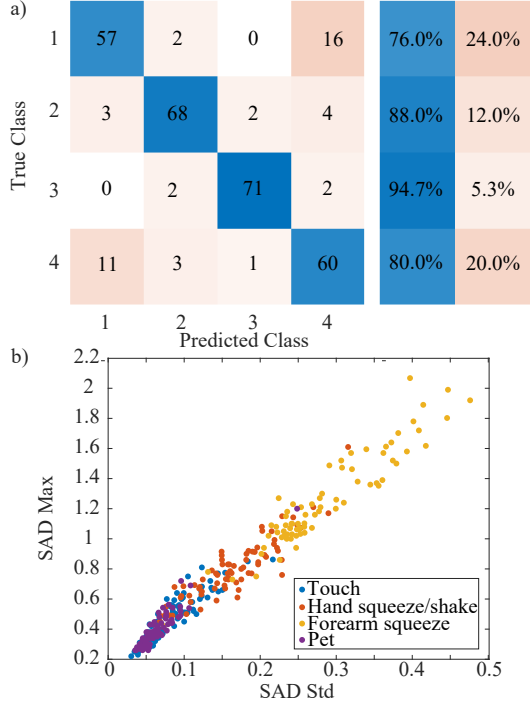


Fig. 5: Classification after combining touches and hand compression interactions. a) Confusion matrix of the cross-validation with $k = 10$. b) Scatter of the two top features.

capturing the distinct intensity variations characteristic of each pHRI type. This qualitative analysis informed the classification architecture, mirroring the methodology of Block *et al.*, who characterized tactile profiles for the HuggieBot. While they identified consistent intra-gesture patterns for “holding” and “patting,” they noted significant morphological variance in “squeeze” profiles across different users [13]. Through a similar tactile profile analysis, the results of this work demonstrate a robust ability to localize and distinguish squeezes; however, the primary challenge remained the differentiation of pets from touches. This ambiguity likely stems from overlapping temporal features, a limitation that could be mitigated by expanding the dataset to include more participants and a higher volume of interaction samples.

In comparison to previous work utilizing Velostat-based sensors, such as the CNN-based object recognition proposed by Yuan *et al.* [8], the overall accuracy of 84.67% achieved in this study is lower than their reported 98.54%. In contrast, this application involves dynamic, high-variance inputs across multiple subjects on a complex cylindrical geometry. While this structure introduced additional noise and signal complexity, it successfully rendered the robotic arm fully sensitive to pHRI for the low-cost SAR.

Moreover, beyond accuracy differences, CNNs generally in-

cur high computational and memory costs when implemented in real-world robotic systems. CNNs often require large numbers of arithmetic operations and substantial model parameters, which can lead to longer processing times and the need for specialized hardware (e.g., GPUs or FPGAs) to achieve real-time performance. This complexity has been noted as a challenge in deploying CNNs for time-sensitive robotic tasks, particularly on systems with limited on-board resources [21], [22].

In this context, the use of an RF classifier was motivated not only by its robustness to noise and intersubject variability but also by its relatively lower computational overhead for the heterogeneous statistical, temporal, and spatial features extracted from tactile signals, as well as in Velostat sensors [17]. RF reduces overfitting and improves generalization compared to single trees by averaging predictions across an ensemble, making them well-suited for tactile classification tasks with complex interaction patterns [23]. Such ensemble approaches have also been shown to perform effectively in robotic tactile and texture classification tasks where complex time-series features and variable interaction patterns are present, providing acceptable accuracy across classes with nonlinear boundaries [24], [25].

In terms of specific hand pHRI classification, Gaitán-Padilla *et al.* in a previous work with the CASTOR robot, developed an FBG-based matrix to identify nine types of pHRI, achieving an accuracy of 98%. Moreover, even when they presented a more sensitive single sensor, they had a limited sensing area of a 2×2 sensing matrix, and some interactions could have risks of missing off-center interactions, for which the Velostat sensor represent a complete cylindrical solution to give a complete surface sensed area [16]. Other social robots have been adapted with soft and flexible sensors, O’Connell *et al.* [14] instrumented a mouse-like soft robot with capacitive sensors. They evaluate the detection of types (5) and locations (3) of pHRI, obtaining an overall accuracy of 83.33%, a result closely aligned with our optimized classification performance (84.67%). This comparison suggests that while soft, flexible sensing arrays may introduce more signal noise than localized high-precision sensors, they provide a more robust and scalable solution for SARs.

Despite the successful classification of pHRI, some limitations inherent to the sensing material and experimental setup must be acknowledged. First, the piezoresistive nature of Velostat introduces hysteresis and non-linear temporal degradation, where the sensor response may not return to baseline immediately following high-force interactions like squeezes. While the use of SAD-based temporal features partially mitigated these effects by focusing on the rate of change rather than absolute magnitude, long-term sensor drift remains a challenge for continuous real-time applications.

Second, the cylindrical geometry of the robotic arm introduces non-uniform mechanical tension across the sensor array. This curvature can lead to pre-loading of certain sensing cells,

potentially skewing the intensity distribution and contributing to the class overlap observed between pet and touch interactions. Finally, this study relied on a controlled dataset with a limited number of participants. Given that the boundaries of the hand and forearm are subjectively defined by the users, the observed intersubject variability suggests that a more diverse group of participants is needed to further generalize the classification model for broader SAR deployment with children.

V. CONCLUSION

This study demonstrated the implementation and evaluation of a low-cost, Velostat-based tactile sensor array integrated onto the CASTOR robot arm for pHRI classification. Through the analysis of spatial heatmaps and temporal features (SAD), the system distinguished between various interaction types, achieving a classification accuracy of 84.67%. While the cylindrical geometry of the arm and the subjective nature of human touch introduced significant signal complexity, the sensor proved highly effective in localizing contact and discriminating between different force intensities. These results suggest that high-density tactile coverage can be achieved without the need for expensive, rigid sensing hardware. Future work will focus on expanding the dataset to improve the classification of low-intensity dynamic interactions, such as "petting," and integrating this tactile feedback into real-time control loops to enhance the social responsiveness of the CASTOR robot.

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